

Al-Aqsa University
Faculty of Computers and IT
Dept. of Info & Computer science



جامعة الأقصى
كلية الحاسبات وتكنولوجيا المعلومات
قسم علم الحاسوب والمعلومات

Operating Systems for Intelligent Systems (ENGS2306)

Final Project Assignment

Automating the IoT Monitoring Hub Deployment

Presented to : **Eng. Mohammed AL-Hams**

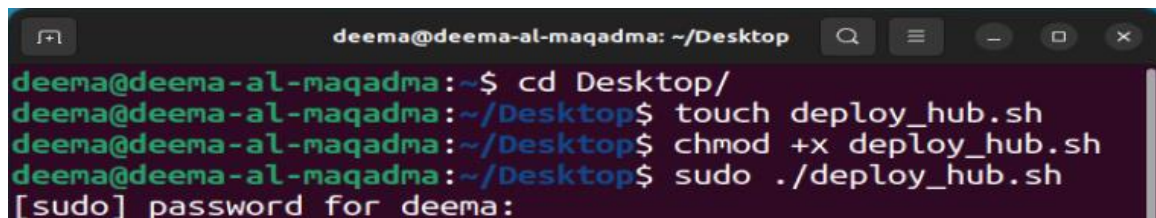
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Student number: : **2320230766**

Semester: Second Semester 2024/2025

1. Project Objective:

This project aims to automate the deployment of an IoT Monitoring Hub using a Bash script on a Linux-based system. The script configures user permissions, sets up a secure web server with Nginx, and creates a simple dashboard to monitor system status. The goal is to simplify and standardize the setup process, ensuring consistency, security, and efficiency across IoT environments.

2. A screenshot of the command created the (deploy_hub.sh) script :



```
deema@deema-al-maqadma: ~/Desktop
deema@deema-al-maqadma:~$ cd Desktop/
deema@deema-al-maqadma:~/Desktop$ touch deploy_hub.sh
deema@deema-al-maqadma:~/Desktop$ chmod +x deploy_hub.sh
deema@deema-al-maqadma:~/Desktop$ sudo ./deploy_hub.sh
[sudo] password for deema:
```

3.The full text of (deploy_hub.sh) fully-commented script:

```
#!/bin/bash

# =====
# --- >IoT Hub Deployment Script
# --- >BY: Deema Mohammed AL-Maqadma
# --- >ID: 2320230766
# --- >TO: Eng. Mohammed AL-Hams
# --- >Automates setup of a secure web-based monitoring hub
# =====

# Requirement 1: Error Handling - Exit immediately if any command fails
set -e

# Function to print colored output for better visibility
print_status() {
    local color_code=""
    case "$1" in
        "red") color_code="31" ;; # Error messages
        "green") color_code="32" ;; # Success messages
        *) color_code="37" ;; # Default white
    esac
    echo -e "\033[0;${color_code}m${2}\033[0m"
}

# Check if script is run with sudo/root privileges - Must run with sudo
if ss -ltn | grep ':80' >/dev/null; then
    print_status "red" "ERROR: Please run this script with sudo: sudo $0"
    exit 1
fi
```

```
print_status "green" "Starting IoT Hub deployment..."

# =====
# Requirement 2: User & Group Management
# =====

# Create hub_admins group
groupadd -f hub_admins
print_status "green" "Created hub_admins group"

# Create iot_admin user with home directory and bash shell
if ! id "iot_admin" &>/dev/null; then
    useradd -m -s /bin/bash iot_admin
    print_status "green" "Created iot_admin user"
else
    print_status "green" "iot_admin user already exists"
fi

# Add user to hub_admins group
usermod -aG hub_admins iot_admin
print_status "green" "Added iot_admin to hub_admins group"

# =====
# Requirement 3: Directory & Permissions Setup
# =====

# Create directory for IoT hub at /srv/iot_hub
mkdir -p /srv/iot_hub
print_status "green" "Created /srv/iot_hub directory"

# Set ownership to iot_admin:hub_admins
chown iot_admin:hub_admins /srv/iot_hub

# Set permissions to 775 (rwx for owner/group, rx for others)
chmod 775 /srv/iot_hub
print_status "green" "Set directory ownership and permissions"

# =====
# Requirement 4: Software Installation
# =====

# Update package lists from repositories
apt update
print_status "green" "Updated package lists"

# Install nginx web server non-interactively
apt install nginx -y
print_status "green" "Installed nginx"

# =====
# Requirement 5: Firewall Configuration
# =====

# Reset firewall to default state
ufw --force reset
```

```

# Set default policies: deny incoming, allow outgoing
ufw default deny incoming
ufw default allow outgoing

# Allow SSH (port 22) and HTTP (port 80) traffic
ufw allow ssh
ufw allow http

# Enable firewall non-interactively (auto-approve)
echo "y" | ufw enable
print_status "green" "Configured firewall (SSH and HTTP allowed)"

# =====

# Requirement 6: Content Deployment
# =====

# Create HTML content with required message
echo "<h1>IoT Hub Dashboard - System Normal</h1>" > /srv/iot_hub/index.html
echo "<p>Last updated: $(date)</p>" >> /srv/iot_hub/index.html
print_status "green" "Created HTML content"

# Configure nginx to serve from custom directory instead of default
# Using sed to replace the root directory in nginx config file
sed -i 's|root /var/www/html;|root /srv/iot_hub;|' /etc/nginx/sites-available/default
print_status "green" "Configured nginx to serve from /srv/iot_hub"

# Test nginx configuration to ensure no errors
if nginx -t; then
    print_status "green" "Nginx configuration test passed"
else
    print_status "red" "Nginx configuration test failed"
    exit 1
fi

# =====

# Requirement 7: Service Management
# =====

# Enable nginx to start automatically on system boot
systemctl enable nginx
print_status "green" "Enabled nginx to start on boot"

# Restart nginx to apply configuration changes
systemctl restart nginx
print_status "green" "Restarted nginx service"

# Wait a moment for nginx to fully start
sleep 8

# =====

# Requirement 8: Final Verification
# =====

print_status "green" "Performing final verification..."

```

```
# Check if nginx process is running
if pgrep nginx >/dev/null; then
    print_status "green" "Nginx process is running"

# Additional check if nginx is listening on port 80
if netstat -ltn | grep ':80' >/dev/null; then
    print_status "green" "Nginx is listening on port 80"

# Test the web page locally with timeout
print_status "green" "Testing web page locally..."

# Try multiple times in case nginx needs more time
for i in {1..3}; do
    if curl -s -f http://localhost >/dev/null 2>&1; then
        print_status "green" "SUCCESS: IoT Hub deployment completed successfully!"
        print_status "green" "Page content:"
        curl -s http://localhost
        print_status "green" "Deployment completed at: $(date)"
        print_status "green" "You can access your IoT Hub at: http://localhost"
        exit 0
    else
        print_status "yellow" "Attempt $i: Web server not ready yet, waiting..."
        sleep 2
    fi
done

# If we get here, curl failed all attempts
print_status "red" "ERROR: Web server is not responding correctly"
print_status "green" "Checking nginx status..."
systemctl status nginx --no-pager -l | head -10
print_status "green" "Checking nginx error logs..."
tail -10 /var/log/nginx/error.log 2>/dev/null || echo "No error logs found"
exit 1

else
    print_status "red" "ERROR: nginx is not listening on port 80"
    print_status "green" "Checking nginx status..."
    systemctl status nginx --no-pager -l | head -10
    exit 1
fi
else
    print_status "red" "ERROR: nginx process is not running"
    print_status "green" "Checking nginx status..."
    systemctl status nginx --no-pager -l | head -10
    exit 1
fi
```

4. A section-by-section explanation of what script does and why:

1. Error Handling:

set -e

Purpose: Makes the script exit immediately if any command fails, preventing continuation with errors.

2. User & Group Management:

groupadd -f hub_admins

useradd -m -s /bin/bash iot_admin

usermod -aG hub_admins iot_admin

Purpose: Creates the hub_admins group, iot_admin user with home directory and bash shell, and adds the user to the group.

3. Directory & Permissions Setup:

mkdir -p /srv/iot_hub

chown iot_admin:hub_admins /srv/iot_hub

chmod 775 /srv/iot_hub

Purpose: Creates the /srv/iot_hub directory and sets ownership to iot_admin:hub_admins with permissions 775 (full control for owner/group, read/execute for others).

4. Software Installation:

apt update

apt install nginx -y

Purpose: Updates package lists and installs the nginx web server non-interactively.

5. Firewall Configuration:

ufw --force reset

ufw default deny incoming

ufw default allow outgoing

ufw allow ssh

ufw allow http

echo "y" | ufw enable

Purpose: Configures the firewall to deny all incoming traffic by default, allow outgoing traffic, and explicitly allow SSH (port 22) and HTTP (port 80) traffic.

6. Content Deployment:

echo "<h1>IoT Hub Dashboard - System Normal</h1>" >

/srv/iot_hub/index.html

echo "<p>Last updated: \$(date)</p>" >> /srv/iot_hub/index.html

sed -i 's|root /var/www/html;|root /srv/iot_hub;|' /etc/nginx/sites-available/default

Purpose: Creates the required HTML content and configures nginx to serve files from the custom directory instead of the default location.

7. Service Management:

systemctl enable nginx

systemctl restart nginx

sleep 8

Purpose: Enables nginx to start automatically on system boot and restarts it to apply configuration changes.

8. Final Verification:

```
if pgrep nginx >/dev/null; then  
    curl -s http://localhost
```

Purpose: Checks if nginx is running and tests the web page locally using curl to verify successful deployment.

--- > This script fulfills all project requirements in a clean, organized manner with proper error handling and verification steps.

5. A screenshot of the command `curl http://localhost` running in the terminal and showing your custom HTML message:

```
#=====
# Requirement 6: Content Deployment
#=====

# Create HTML content with required message
echo "<h1>IoT Hub Dashboard- System Normal</h1>" > /srv/iot_hub/index.html
echo "<p>Last updated: $(date)</p>" >> /srv/iot_hub/index.html
print_status "green" "---> Created HTML content"

# Configure nginx to serve from custom directory instead of default
# Using sed to replace the root directory in nginx config file
sed -i 's|root /var/www/html;|root /srv/iot_hub;|' /etc/nginx/sites-available/default
print_status "green" "---> Configured nginx to serve from /srv/iot_hub"
#systemctl reload nginx
```

```
deema@deema-al-maqadma: ~/Desktop
---> You can access your IoT Hub at: http://localhost
deema@deema-al-maqadma:~/Desktop$
```

* Expected Output :

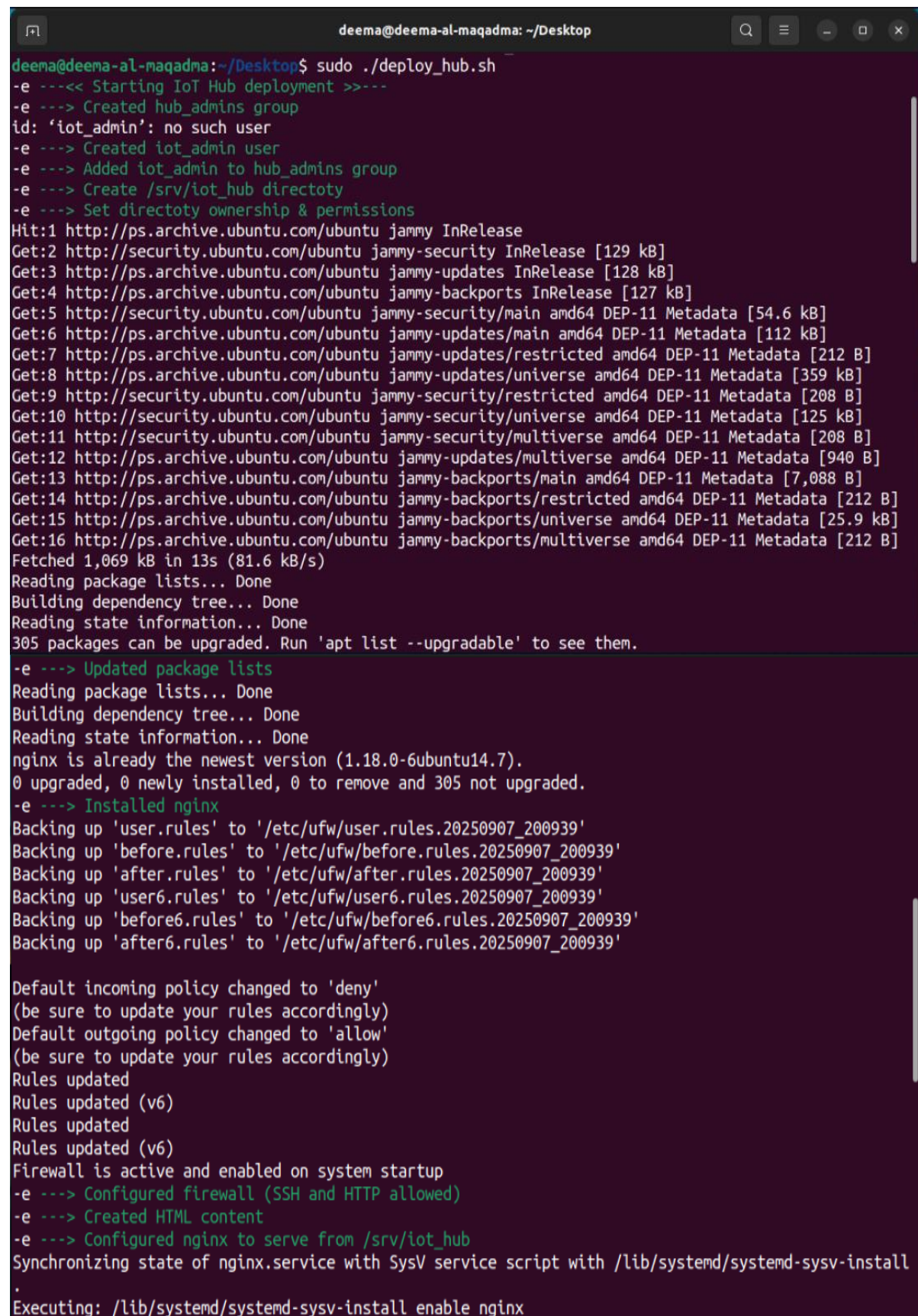
```
<h1>IoT Hub Dashboard- System Normal</h1> <p>Last updated: Mon Sep 8 02:52:53 PM EEST 2025</p>
deema@deema-al-maqadma:~/Desktop$
```

```
deema@deema-al-maqadma:~/Desktop$ curl http://localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
    body {
        width: 35em;
        margin: 0 auto;
        font-family: Tahoma, Verdana, Arial, sans-serif;
    }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
deema@deema-al-maqadma:~/Desktop$
```

6. A screenshot of the script running successfully from the terminal:



```
deema@deema-al-maqadma: ~/Desktop
deema@deema-al-maqadma:~/Desktop$ sudo ./deploy_hub.sh
-e ---<< Starting IoT Hub deployment >>---
-e ---> Created hub_admins group
id: 'iot_admin': no such user
-e ---> Created iot_admin user
-e ---> Added iot_admin to hub_admins group
-e ---> Create /srv/iot_hub directoty
-e ---> Set directoty ownership & permissions
Hit:1 http://ps.archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:3 http://ps.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:4 http://ps.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:5 http://security.ubuntu.com/ubuntu jammy-security/main amd64 DEP-11 Metadata [54.6 kB]
Get:6 http://ps.archive.ubuntu.com/ubuntu jammy-updates/main amd64 DEP-11 Metadata [112 kB]
Get:7 http://ps.archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 DEP-11 Metadata [212 B]
Get:8 http://ps.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 DEP-11 Metadata [359 kB]
Get:9 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 DEP-11 Metadata [208 B]
Get:10 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 DEP-11 Metadata [125 kB]
Get:11 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 DEP-11 Metadata [208 B]
Get:12 http://ps.archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 DEP-11 Metadata [940 B]
Get:13 http://ps.archive.ubuntu.com/ubuntu jammy-backports/main amd64 DEP-11 Metadata [7,088 B]
Get:14 http://ps.archive.ubuntu.com/ubuntu jammy-backports/restricted amd64 DEP-11 Metadata [212 B]
Get:15 http://ps.archive.ubuntu.com/ubuntu jammy-backports/universe amd64 DEP-11 Metadata [25.9 kB]
Get:16 http://ps.archive.ubuntu.com/ubuntu jammy-backports/multiverse amd64 DEP-11 Metadata [212 B]
Fetched 1,069 kB in 13s (81.6 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
305 packages can be upgraded. Run 'apt list --upgradable' to see them.
-e ---> Updated package lists
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nginx is already the newest version (1.18.0-6ubuntu14.7).
0 upgraded, 0 newly installed, 0 to remove and 305 not upgraded.
-e ---> Installed nginx
Backing up 'user.rules' to '/etc/ufw/user.rules.20250907_200939'
Backing up 'before.rules' to '/etc/ufw/before.rules.20250907_200939'
Backing up 'after.rules' to '/etc/ufw/after.rules.20250907_200939'
Backing up 'user6.rules' to '/etc/ufw/user6.rules.20250907_200939'
Backing up 'before6.rules' to '/etc/ufw/before6.rules.20250907_200939'
Backing up 'after6.rules' to '/etc/ufw/after6.rules.20250907_200939'

Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
Firewall is active and enabled on system startup
-e ---> Configured firewall (SSH and HTTP allowed)
-e ---> Created HTML content
-e ---> Configured nginx to serve from /srv/iot_hub
Synchronizing state of nginx.service with SysV service script with /lib/systemd/systemd-sysv-install
.
Executing: /lib/systemd/systemd-sysv-install enable nginx
```

```
-e ---> Enabled nginx to start boot
-e ---> Restarted nginx to service
-e ---> Performing Final Verification
-e ---> Nginx process is running
-e ---> Nginx is listening on port 80
-e ---> Test web page locally...
-e ---> SUCCESS: IoT Hub deployment completed successfully!
Page content:
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
  body {
    width: 35em;
    margin: 0 auto;
    font-family: Tahoma, Verdana, Arial, sans-serif;
  }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>
```

```
deema@deema-al-maqadma: ~/Desktop
-e ---> SUCCESS: IoT Hub deployment completed successfully!
Page content:
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
  body {
    width: 35em;
    margin: 0 auto;
    font-family: Tahoma, Verdana, Arial, sans-serif;
  }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
-e ---> Deployment completed at: Sun Sep  7 08:09:52 PM EEST 2025
-e ---> You can access your IoT Hub at: http://localhost
deema@deema-al-maqadma:~/Desktop$
```

7. Script Improvements Explanation:

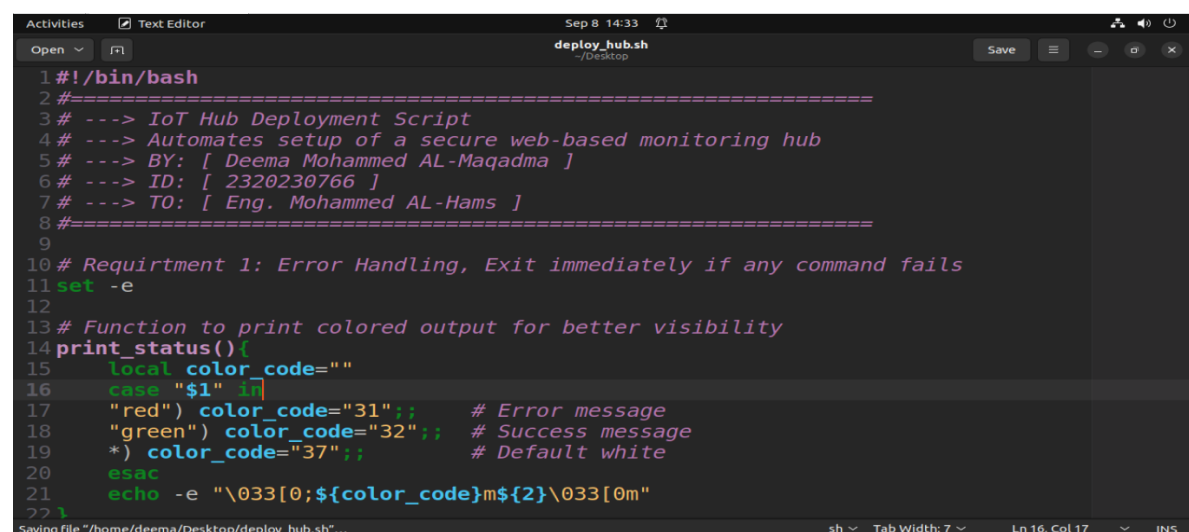
To enhance the script's performance and readability, several improvements were implemented:

- **print_status()**: A custom function was added to print colored messages in the terminal. This improves user experience by visually distinguishing success, warnings, and errors, making the script output more intuitive and easier to debug.
 - **sleep 8**: This line introduces a short delay between operations to ensure system stability and allow background processes to complete before proceeding. It improves reliability, especially during service restarts or file operations.
 - **curl -s http://localhost**: This command tests the availability of the local web server after deployment. It ensures that the Nginx service is running correctly and the interface is accessible, providing a quick validation step.
-

8. Report Explanation :

> This report documents the process of automating the deployment of an IoT Monitoring Hub using a Bash script on a Linux-based system. It explains each step of the script, including user and group setup, directory configuration, firewall rules, and web server deployment using Nginx. The report also includes screenshots of the execution process and the resulting web interface. The goal is to provide a clear and reproducible method for setting up a secure and functional IoT hub environment.

9. A screenshot of the full (deploy_hub.sh) script in Linux:



```
1#!/bin/bash
2#=====
3# ---> IoT Hub Deployment Script
4# ---> Automates setup of a secure web-based monitoring hub
5# ---> BY: [ Deema Mohammed AL-Maqadma ]
6# ---> ID: [ 2320230766 ]
7# ---> TO: [ Eng. Mohammed AL-Hams ]
8#=====
9
10# Requirement 1: Error Handling, Exit immediately if any command fails
11set -e
12
13# Function to print colored output for better visibility
14print_status(){
15    local color_code=""
16    case "$1" in
17        "red") color_code="31";; # Error message
18        "green") color_code="32";; # Success message
19        *) color_code="37";; # Default white
20    esac
21    echo -e "\033[0;${color_code}m${2}\033[0m"
22 }
```

Saving file "/home/deema/Desktop/deploy_hub.sh"...

sh Tab Width: 7 Ln 16, Col 17 INS

```

24 # Check if script is run with sudo/root privileges (must run with sudo)
25 if [ "$(whoami)" != "root" ]; then
26     print_status "red" "ERROR: Please run this script with sudo: sudo $0"
27     exit 1
28 fi
29 print_status "green" "----<< Starting IoT Hub deployment >>----"
30
31 #=====
32 # Requirement 2: User & Group Management
33 #=====
34
35 # Create hub_admins group
36 groupadd -f hub_admins
37 print_status "green" "----> Created hub_admins group"
38
39 # Create iot_admin user with home directory and bash shell
40 if ! id "iot_admin" &>/dev/null; then
41     useradd -m -s /bin/bash iot_admin
42     print_status "green" "----> Created iot_admin user"
43 else
44     print_status "green" "----> iot_admin user already exists"
45 fi
46
47 # Add user to hub_admins group
48 usermod -aG hub_admins iot_admin
49 print_status "green" "----> Added iot_admin to hub_admins group"
50
51 #=====
52 # Requirement 3: Directory & Permissions Setup
53 #=====
54
55 # Create directoty for IoT Hub at /srv/iot_hub
56 mkdir -p /srv/iot_hub
57 print_status "green" "----> Create /srv/iot_hub directoty"
58
59 # Set ownership to iot admin:hub admins
60 chown iot_admin:hub_admins /srv/iot_hub
61
62 # Set Permissions to 775 (rwx for owner/group , rw for others)
63 chmod 775 /srv/iot_hub
64 print_status "green" "----> Set directoty ownership & permissions"
65
66 #=====
67 # Requirement 4: Software Installation
68 #=====
69
70 # Update package lists from repositories
71 apt update
72 print_status "green" "----> Updated package lists"
73
74 # install nginx web server non-interactively
75 apt install nginx -y
76 print_status "green" "----> Installed nginx"
77
78 #=====
79 # Requirement 5: Firewall Configuration
80 #=====
81
82 # Reset firewall to default state
83 ufw --force reset
84
85 # Set default policies: deny incoming, allow outgoing
86 ufw default deny incoming
87 ufw default allow outgoing
88
89 # Allow SSH (port 22) and HTTP (port 80) traffic
90 ufw allow ssh
91 ufw allow http
92
93 # Enable firewall non-interactively (auto-approve)
94 echo "y" | ufw enable
95 print_status "green" "----> Configured firewall (SSH and HTTP allowed)"
96
97 #=====
98 # Requirement 6: Content Deployment
99 #=====
100
101 # Create HTML content with required message
102 echo "<h1>IoT Hub Dashboard- System Normal</h1>" > /srv/iot_hub/index.html
103 echo "<p>Last updated: $(date)</p>" >> /srv/iot_hub/index.html
104 print_status "green" "----> Created HTML content"
105
106 # Configure nginx to serve from custom directory instead of default
107 # Using sed to replace the root directory in nginx config file
108 sed -i 's|root /var/www/html;|root /srv/iot_hub;|' /etc/nginx/sites-available/default
109 print_status "green" "----> Configured nginx to serve from /srv/iot_hub"
110 #systemctl reload nginx
111
112 #=====
113 # Requirement 7: Service Management
114 #=====

```

```

116 # Test nginx configuration first to ensure no errors
117 if nginx -t; then
118     print_status "green" "---> Nginx configuration test passed"
119 else
120     print_status "red" "---> Nginx configuration test failed"
121     exit 1
122 fi
123
124 # Enable nginx to start automatically on system boot
125 systemctl enable nginx
126 print_status "green" "---> Enabled nginx to start boot"
127
128 # Restart nginx to apply configuration changes
129 systemctl restart nginx
130 print_status "green" "---> Restarted nginx to service"
131
132 # Wait a moment for nginx to fully start
133 sleep 8
134 #=====
135 # Requirement 8: Final Verification
136 #=====
137
138
139 print_status "green" "---> Rerforming Final Verification"
140
141 # Check if nginx process is running using pgrep
142 if ! pgrep nginx >/dev/null; then
143     print_status "green" "---> Nginx process is running "
144
145     # Additional check if nginx is listening on port 80
146     if ss -ltn | grep ':80' >/dev/null; then
147         print_status "green" "---> Nginx is listening on port 80"
148
149         # Test the web page locally with timeout
150         print_status "green" "---> Test web page locally..."
151
152         # Try multiple time in case nginx needs more time
153         for i in {1..3}; do
154             if curl -s -f http://localhost>/dev/null 2>&1; then
155                 print_status "green" "---> SUCCESS: IoT Hub deployment compleated
156                 successfully!"
157                 echo "Page content:"
158                 curl -s http://localhost
159                 print_status "green" "---> Deployment completed at: ${date}"
160                 print_status "green" "---> You can access your IoT Hub at: http://
161                 localhost"
162                 exit 0
163             else
164                 print_status "yellow" "---> Attempt $i: Web server not ready yet,
165                 waiting..."
166                 sleep 2
167             fi
168         done
169
170         # If we get here, curl failed all attempts
171         print_status "red" "---> ERROR: Web server is not responding correctly"
172         print_status "green" "---> Checking nginx status..."
173         systemctl status nginx --no-pager -l | head -n 10
174         print_status "green" "---> Checking nginx error logs..."
175         tail -10 /var/log/nginx/error.log 2>/dev/null || echo "No error logs found"
176         exit 1
177     else
178         print_status "red" "---> ERROR: Nginx is not listening on port 80"
179         print_status "green" "---> Checking nginx status..."
180         systemctl status nginx --no-pager -l | head -n 10
181         exit 1
182     fi
183 else
184     print_status "red" "---> ERROR: Nginx process is not runinnng"
185     print_status "green" "---> Checking nginx status..."
186     systemctl status nginx --no-pager -l | head -n 10
187     exit 1
188 fi
189
190 print_status "green" "---> Deployment completed at: ${date}"
191 print_status "green" "---> You can access your IoT Hub at: http://localhost"
192

```

"واخر دعواهم ان الحمد لله رب العالمين ."